AArch64 Memory & Cache System — Technical Reference Workbook (Based On arm64 Cortex A72) 1
Chapter 2 — Architectural Configuration of the Memory System (MAIR, TCR, TTBR, SCTLR) 1
2.1 Scope 1
2.2 MAIR_EL1 — Memory Attribute Indirection Register 1
2.2.1 Structure 1
2.2.2 Common Encodings 2
Note: These 3 encodings are for understanding. There is other encoding also possible based on device type. To understand in detail please check out my earlier MMU post. 2
2.2.3 Architectural Impact 2
2.3 TCR_EL1 — Translation Control Register 2
2.3.1 Key Fields 3
2.3.2 Architectural Impact 3
2.4 TTBR0_EL1 / TTBR1_EL1 — Translation Table Base Registers 3
2.4.1 TTBR0_EL1 3
2.4.2 TTBR1_EL1 4
2.4.3 Architectural Impact 4
2.5 SCTLR_EL1 — System Control Register 4
2.5.1 Key Fields 4
2.5.2 Architectural Impact 4
2.6 Required Initialization Sequence 5
2.7 Exercises (Technical Reference) 5
Exercise 2.1 — Attribute Encoding 5
Exercise 2.2 — Shareability 5
Exercise 2.3 — TTBR Configuration 6
Exercise 2.4 — SCTLR Interaction 6

AArch64 Memory & Cache System — Technical Reference Workbook (Based On arm64 Cortex A72)

Chapter 2 — Architectural Configuration of the Memory System (MAIR, TCR, TTBR, SCTLR)

2.1 Scope

This chapter defines the architectural registers that configure the AArch64 memory system:

- **MAIR_EL1** — Memory Attribute Indirection Register
- **TCR_EL1** — Translation Control Register
- **TTBR0_EL1 / TTBR1_EL1** — Translation Table Base Registers
- **SCTLR_EL1** — System Control Register

Correct configuration of these registers is required before enabling the MMU and caches, and before performing any cache maintenance operations.

2.2 MAIR_EL1 — Memory Attribute Indirection Register

MAIR_EL1 defines the memory attribute encodings used by translation tables.

2.2.1 Structure

MAIR_EL1 contains eight 8-bit fields:

```
MAIR_EL1[7:0]   → Attr0
MAIR_EL1[15:8]  → Attr1
...
MAIR_EL1[63:56] → Attr7
```

Each field defines:

- Cacheability
- Inner/outer policy
- Device vs Normal memory
- Write-back / write-through / non-cacheable behavior

2.2.2 Common Encodings

Attribute	Meaning	Typical Encoding
Normal, Inner/Outer Write-Back, Read-Allocate, Write-Allocate	Fully cacheable	`0xFF`
Normal, Non-Cacheable	No caching	`0x44`
Device-nGnRnE	Strongly ordered device	`0x00`

Note: These 3 encodings are for understanding. There is other encoding also possible based on device type. To understand in detail please check out my earlier MMU post.

2.2.3 Architectural Impact

MAIR_EL1 determines:

- Whether DC/IC operations have architectural effect
- Whether dirty lines may exist
- Whether PoU/PoC semantics apply
- Whether speculative fetch is permitted

Incorrect MAIR configuration results in undefined cache behavior.

2.3 TCR_EL1 — Translation Control Register

TCR_EL1 defines:

- Address size
- Granule size
- Shareability
- Cacheability
- Physical address size

- TTBR0/TTBR1 region split
-

2.3.1 Key Fields

Field	Purpose
TCR_EL1.T0SZ	Size of TTBR0 region
TCR_EL1.T1SZ	Size of TTBR1 region
TCR_EL1.IRGN0 / ORGN0	Inner/outer cacheability for TTBR0
TCR_EL1.IRGN1 / ORGN1	Inner/outer cacheability for TTBR1
TCR_EL1.SH0 / SH1	Shareability (Non-shareable / Inner / Outer)
TCR_EL1.TG0 / TG1	Granule size (4KB, 16KB, 64KB)
TCR_EL1.PS	Physical address size

2.3.2 Architectural Impact

TCR_EL1 determines:

- Whether memory regions are inner-shareable
- Whether L1/L2 caches participate in coherency
- Whether PoU and PoC are meaningful
- Whether speculative access is permitted

Incorrect TCR configuration can break coherency and invalidate cache maintenance behavior.

2.4 TTBR0_EL1 / TTBR1_EL1 — Translation Table Base Registers

These registers define the base addresses of the translation tables.

2.4.1 TTBR0_EL1

Used for lower VA range (typically user space or bootloader mappings).

2.4.2 TTBR1_EL1

Used for higher VA range (typically kernel mappings).

2.4.3 Architectural Impact

TTBR registers determine:

- Which memory regions are mapped
- Which attributes apply to each region
- Whether DC/IC operations target the correct physical memory
- Whether MMIO regions are correctly marked Device-nGnRnE

Incorrect TTBR configuration results in:

- Cache maintenance targeting wrong physical addresses
- MMIO regions being cached
- DMA buffers being incorrectly mapped

2.5 SCTLR_EL1 — System Control Register

SCTLR_EL1 controls global memory system behavior.

2.5.1 Key Fields

Field	Meaning
SCTLR_EL1.M	MMU enable
SCTLR_EL1.C	Data cache enable
SCTLR_EL1.I	Instruction cache enable
SCTLR_EL1.SA0 / SA	Stack alignment
SCTLR_EL1.A	Alignment checking
SCTLR_EL1.WXN	Write-xor-execute enforcement

2.5.2 Architectural Impact

Enabling SCTLR_EL1.M:

- Activates MMU
- Activates program-flow prediction (branch prediction)

- Activates attribute system (MAIR/TCR/TTBR)
- Makes PoU/PoC meaningful

Enabling SCTLR_EL1.C and SCTLR_EL1.I:

- Activates L1D and L1I
- Allows dirty lines to exist
- Allows instruction caching
- Requires correct DC/IC usage

2.6 Required Initialization Sequence

The architectural sequence for correct memory system initialization is:

1. **Disable MMU and caches** `SCTLR_EL1.M = 0`, `SCTLR_EL1.C = 0`, `SCTLR_EL1.I = 0`
2. **Configure MAIR_EL1** Define attribute encodings for Normal, Device, Non-cacheable regions.
3. **Configure TCR_EL1** Set shareability, cacheability, granule size, and address size.
4. **Configure TTBR0_EL1 / TTBR1_EL1** Install translation tables with correct attributes.
5. **Optionally configure implementation-specific features** (prefetcher, ACTLR_EL1, prediction restrictions)
6. **Enable MMU** `SCTLR_EL1.M = 1`
7. **Enable caches** `SCTLR_EL1.C = 1`, `SCTLR_EL1.I = 1`
8. **Perform required DC/IC maintenance** As defined in later chapters.

2.7 Exercises (Technical Reference)

Exercise 2.1 — Attribute Encoding

Explain why Device-nGnRnE must be encoded as non-cacheable in MAIR_EL1.

Exercise 2.2 — Shareability

Describe the impact of configuring a Normal memory region as Non-shareable in TCR_EL1.

Exercise 2.3 — TTBR Configuration

Explain how incorrect TTBR0_EL1 configuration can cause DC CVAC to write back to the wrong physical address.

Exercise 2.4 — SCTLR Interaction

Describe the architectural effect of enabling SCTLR_EL1.C before MAIR/TCR/TTBR are configured.